Homework – April 9, 2024

1. Solve $dy/dx = y - x$ with $y(0) = 1$ using the linear multi-step method
   $Y_{m+1} + (3/2)y_m - 3y_{m-1} + (1/2)y_{m-2} = 3hf_m$.

   First, find the exact solution for the given problem. Then, make use of the exact solution for any additional starting values in order to apply the linear multi-step method. Compare your numerical solution against the exact solution. Explain your observation.

2. For the initial value problems given below, find the solutions using the Picard iteration method.

(i) $dy/dx = x\,y$ with $y(0) = 1$, $x \in [0,1]$.

(ii) $dy/dx = 2(y+1)$ with $y(0) = 0$, $x \in [0,1]$.

3. A numerical method known as the Runge-Kutta method is given as $y_{m+1} = y_m + (h/6)(F_1 + 4F_2 + F_3)$, where $F_1 = f(y_m)$, $F_2 = f(y_m + 0.5hF_1)$ and $F_3 = f(y_m - hF_1 + 2hF_2)$. Find the order of this method. Using this method, find the *exact discrete solution* of the initial value problem $dy/dx = -5y$ with $y(0) = 1$.